Ruijie (Jack) Zhang

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RESEARCH PROFILE

PhD researcher integrating single-cell and spatial transcriptomics, quantitative image analysis, and molecular and cellular assays to investigate tissue immune organization and disease-associated cell states.

EDUCATION

University of Illinois Urbana-Champaign — **Champaign, IL**
PhD in Nutritional Sciences — Aug 2023 – Present
Expected graduation: Dec 2027.
Graduate College Mentoring Certificate.

University of Connecticut — **Storrs, CT**
MS in Nutritional Sciences — Aug 2021 – May 2023

Shenzhen University — **Shenzhen, China**
BE in Food Science and Engineering — Sep 2017 – Jun 2021

PEER-REVIEWED PUBLICATIONS

Chen, J., **Zhang, R.**, Zhou, C., et al. (2024). The impact of short-term changes in sleeping and eating patterns on glucometabolic health and gut microbiota in healthy young adults: a proof-of-concept controlled feeding study. *Food Science and Human Wellness*, 13(6), 3553–3569. **Co-first author.**

Andersen, C. J., Huang, L., Zhai, F., Esposito, C. P., Greco, J. M., **Zhang, R.**, … & Van Dyke, A. R. (2023). Consumption of different egg-based diets alters clinical metabolic and hematological parameters in young, healthy men and women. *Nutrients*, 15(17), 3747.

Chen, J. H., Lu, L. W., Ge, Q., Feng, D., Yu, J., Liu, B., **Zhang, R.**, … & Chen, F. (2023). Missing puzzle pieces of time-restricted-eating (TRE) as a long-term weight-loss strategy in overweight and obese people? A systematic review and meta-analysis of randomized controlled trials. *Critical Reviews in Food Science and Nutrition*, 63(15), 2331–2347.

MANUSCRIPT IN PREPARATION

Zhang, R., et al. RNA-binding protein hnRNP I loss in mouse colon epithelial cells disrupts immune defense and leads to hyperproliferation. Manuscript in preparation.

TECHNICAL SKILLS AND LANGUAGES

Bioinformatics: Single-cell RNA-seq and spatial transcriptomics (Seurat, Scanpy); NGS data processing (STARsolo, Cell Ranger, Space Ranger); dimensionality reduction (PCA, UMAP, t-SNE); gene regulatory networks (SCENIC); splice-junction analysis; 16S rRNA gene sequencing analysis.

Experimental methods: Mouse breeding, genotyping, dissection, and intraperitoneal injection; tissue processing and sectioning; H&E, AB-PAS, IHC, and multiplex IF; confocal and epifluorescence microscopy; qPCR; DNA/RNA extraction; Western blotting; ChIP; ELISA; bacterial culture and transformation; cell and organoid culture.

Image analysis: Quantitative microscopy image analysis using Fiji/ImageJ, QuPath, and Python (scikit-image).

Programming and statistics: R (tidyverse, ggplot2, Bioconductor); Python (pandas, NumPy, scikit-learn); generalized linear and linear mixed models; Bash; Git; Linux; HPC workflows and job scheduling (Slurm).

Languages: English (proficient); Mandarin Chinese (native); Cantonese (fluent); Spanish (beginner).

RESEARCH EXPERIENCE

Graduate Research Assistant Aug 2023 – Present
University of Illinois Urbana-Champaign | Division of Nutritional Sciences
Current advisor: Dr. Michael Robben; former advisor: Dr. Yuan-Xiang Pan (deceased).

Dissertation research in the hnRNP I knockout model

- Investigate the role of the RNA-binding protein hnRNP I in intestinal epithelial homeostasis using transcriptomic analyses, tissue phenotyping, and molecular assays.
- Integrate single-cell and spatial transcriptomics to characterize epithelial cell states, regional differentiation, and the organization of colonic immune populations.
- Develop computational workflows in R and Python for cell-state classification, tissue deconvolution, differential expression, gene regulatory network analysis, and splice-junction analysis.
- Identify altered epithelial splicing and reduced JAK–STAT and antimicrobial gene expression programs in hnRNP I-deficient mice. Preparing a first-author manuscript; presented related findings at the American Physiology Summit 2026.

Additional research projects

- Contribute to mouse studies and spatial transcriptomics in a collaborative project investigating immune dysregulation in type 1 diabetes.
- Lead experimental work evaluating the effects of omega-3 fatty acids in AOM/DSS-treated mice.
- Optimize and perform multiplex immunofluorescence (IBEX) staining of human colorectal cancer tissue microarrays to characterize macrophage heterogeneity.

Graduate Research Assistant Aug 2021 – May 2023
University of Connecticut | Department of Nutritional Sciences
Advisor: Dr. Catherine J. Andersen

- Conducted secondary analyses of a randomized dietary intervention trial examining how combined oral contraceptive use modifies metabolic responses to egg consumption in young adults.
- Applied linear mixed models to account for repeated measures and within-participant variability when evaluating HDL lipoprotein subclasses and particle composition.
- Collaborated on data quality control, biological sample processing, and serum metabolic profiling.
- Coauthored a publication in Nutrients and presented findings at ASN NUTRITION 2022 and 2023.

Undergraduate Research Assistant Jul 2020 – Jun 2021
Shenzhen University | Institute of Advanced Study
Advisor: Dr. Jie-Hua Chen

- Coordinated participant recruitment, dietary compliance, and sample collection for a controlled feeding study examining sleeping and eating patterns, glucose metabolism, and the gut microbiome.
- Analyzed 16S rRNA gene sequencing data and metabolic biomarkers to evaluate the effects of circadian disruption and subsequent schedule realignment.
- Published findings as a co-first author in Food Science and Human Wellness.

INTERNSHIP EXPERIENCE

Research and Development Summer Intern May 2020 – Aug 2020
Tyson Foods China (Mickey Food Services Ltd.) | Shenzhen, China

- Contributed to poultry product development for key accounts, including McDonald's Hong Kong.
- Conducted sensory evaluations and supported optimization of food additive formulations for industrial processing.

CONFERENCE PRESENTATIONS

Zhang, R., & Robben, M. (2026). RNA-binding protein hnRNP I loss in mouse colon epithelial cells disrupts innate immune defense and leads to hyperproliferation. Presented at the American Physiology Summit 2026.

Zhang, R., Esposito, C. P., Greco, J., Huang, L., & Andersen, C. (2023). P16-041-23 Combined oral contraceptive use alters the effects of egg intake on lipoprotein profiles, without impacting effects on insulin resistance, inflammation, or choline status. *Current Developments in Nutrition*, 7, 100801. Presented at ASN NUTRITION 2023.

Zhang, R., Esposito, C. P., Greco, J., & Andersen, C. (2022). Consumption of different egg-based diets minimally, yet differentially, impacts markers insulin resistance in young healthy adults. *Current Developments in Nutrition*, 6(Suppl 1), 461. Presented at ASN NUTRITION 2022.

COAUTHORED CONFERENCE ABSTRACTS

Huang, L., Woodruff, R., …, **Zhang, R.**, … & Andersen, C. (2023). P30-015-23 Egg intake alters inflammatory properties of HDL and immune cells under basal and pathogenic conditions, which uniquely correlate with changes in HDL composition. *Current Developments in Nutrition*, 7, 101473.

Zhai, F., Huang, L., …, **Zhang, R.**, … & Andersen, C. J. (2023). P30-041-23 Effects of egg consumption on the capacity of HDL to modulate T cell activation. *Current Developments in Nutrition*, 7, 101499.

Zhai, F., Huang, L., …, **Zhang, R.**, … & Andersen, C. J. (2023). Immunomodulatory properties of HDL are associated with lipidomic and proteomic composition and are modifiable by diet. *Arteriosclerosis, Thrombosis, and Vascular Biology*, 43(Suppl 1), A167.

Esposito, C. P., Andersen, C. J., **Zhang, R.**, & Greco, J. (2023). Combined oral contraception use differentially alters cardiometabolic disease risk factors in response to egg consumption in young, healthy women. American College of Nurse-Midwives, 68th Annual Meeting & Exhibition, Orlando, FL.

Huang, L., Garcia, C., Corvino, O., Woo, H., Jang, H., **Zhang, R.**, … & Andersen, C. J. (2022). Egg consumption induces sex-specific changes in high-density lipoprotein composition and function. *Circulation*, 146(Suppl 1), A14231.

TEACHING AND MENTORING

Graduate Tutor Jun 2026 – Present

University of Illinois Urbana-Champaign | PhD Qualifying Examination Preparation

- Provide 5.5 hours of weekly individual and small-group tutoring in nutrition, biochemistry, and metabolism for doctoral students preparing for the qualifying examination.
- Develop review sessions and coach students in organizing clear responses to committee-style questions.

URAP Research Mentor Jan 2025 – May 2025

University of Illinois Urbana-Champaign | Undergraduate Research Apprenticeship Program

- Mentored an undergraduate researcher in experimental design, qPCR, data analysis, and visualization in R.
- Guided preparation of research findings for the Undergraduate Research Symposium.

Graduate Teaching Assistant Aug 2022 – Jan 2023

University of Connecticut | Human Anatomy and Physiology

- Instructed two weekly laboratory sections (36 students), led Anatomage activities, supervised dissections, and assessed laboratory reports and practical examinations.

AWARDS AND HONORS

University of Illinois Urbana-Champaign, Division of Nutritional Sciences

Margin of Excellence Research Grant 2026

William Rose Endowed Award 2026

Research Excellence Assistantship Award 2023